Assignment: Time Complexity Analysis

Hint: The basic operation for each code has been highlighted in bold. You should use this as the core operation to determine the time complexity.

In this assignment, you will analyse three different C++ algorithms.
For each algorithm, answer the following questions:

1. Time Complexity $T(n)$
   Express the total number of times the basic operation is executed.
2. Maximum Input Size (n)
   Assume the CPU can perform 1 billion operations per second (1 GHz CPU).
   Calculate the largest n that the algorithm can handle within this time frame.

Code 1:

```
  int sum = 0;
  for (int i = 1; i <= n; i++)
    sum += i;
   return sum;
}
```

Code 2:

```
  int sum = 0;
  for (int i = 0; i < n; i++)
    for (int j = i + 1; j < n; j++) {
      sum += arr[i] + arr[j];
  return sum;
```

Code 3:

```
Int Max(int a[], int n) {
  return a[n-1];  // Assume the last element is the maximum (sorted array)
}
```